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AVDASI-3: Sustainable Aviation & Ethics

Lecture-5: Sustainable Structures & Materials

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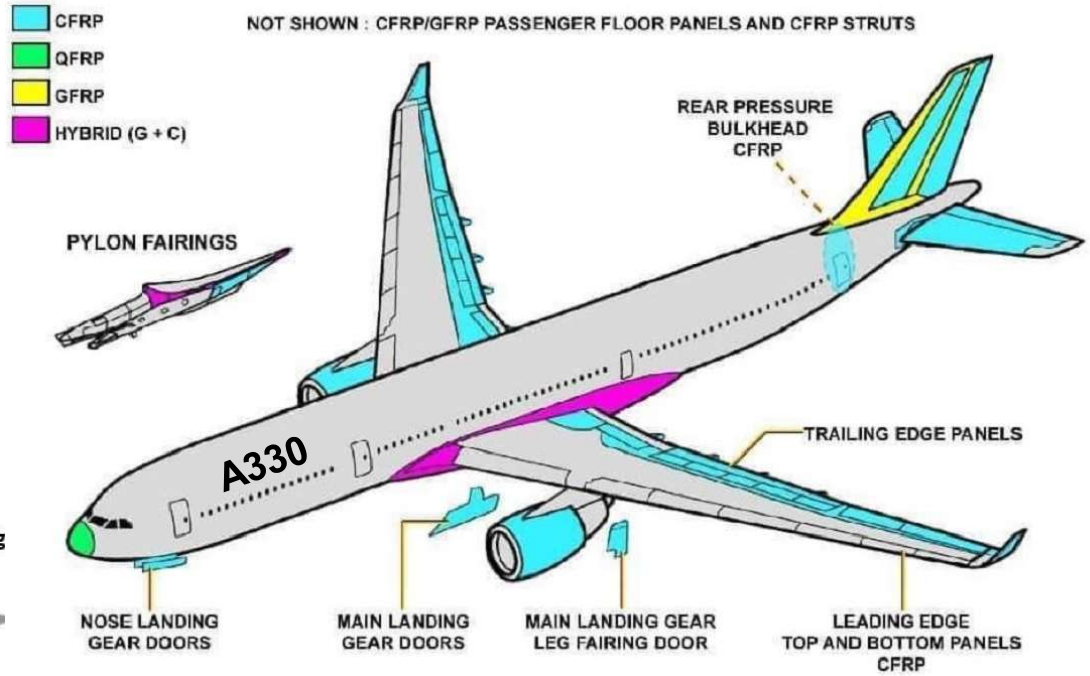
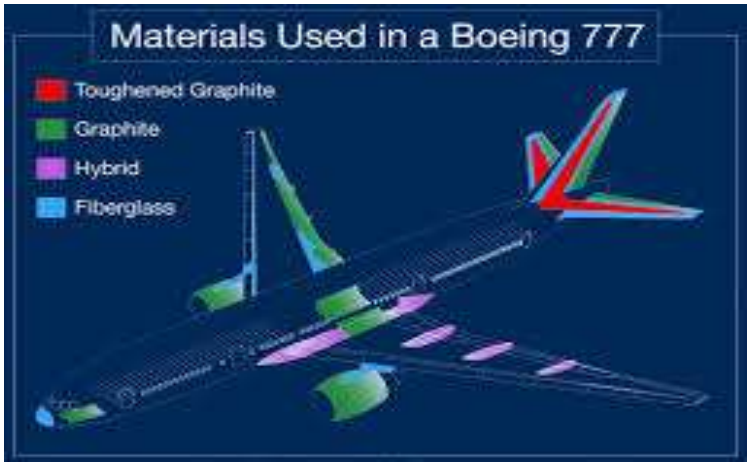
Typical airframe





The inverse snowball promise

To maintain speed, T/O & landing distances, RoC etc.

Can reduce thrust

Weight decreases

From smaller surface area, smaller engines, less fuel etc.

Both lift induced (for lower lift) and skin friction (for lower surface area)

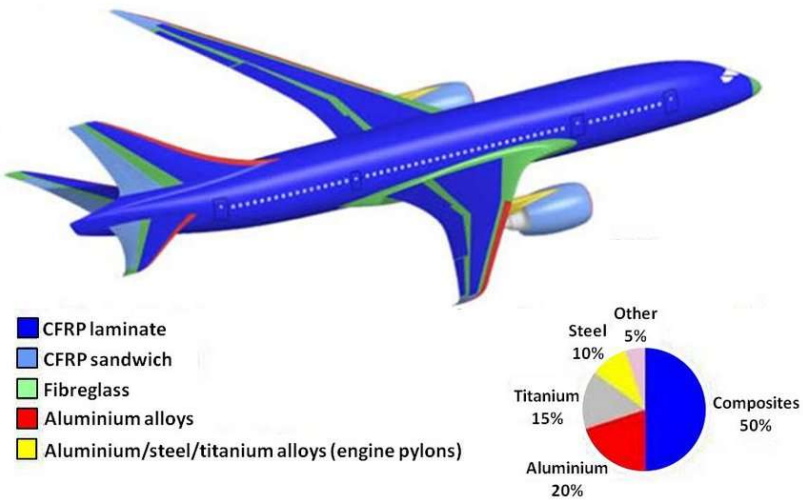
Drag reduces

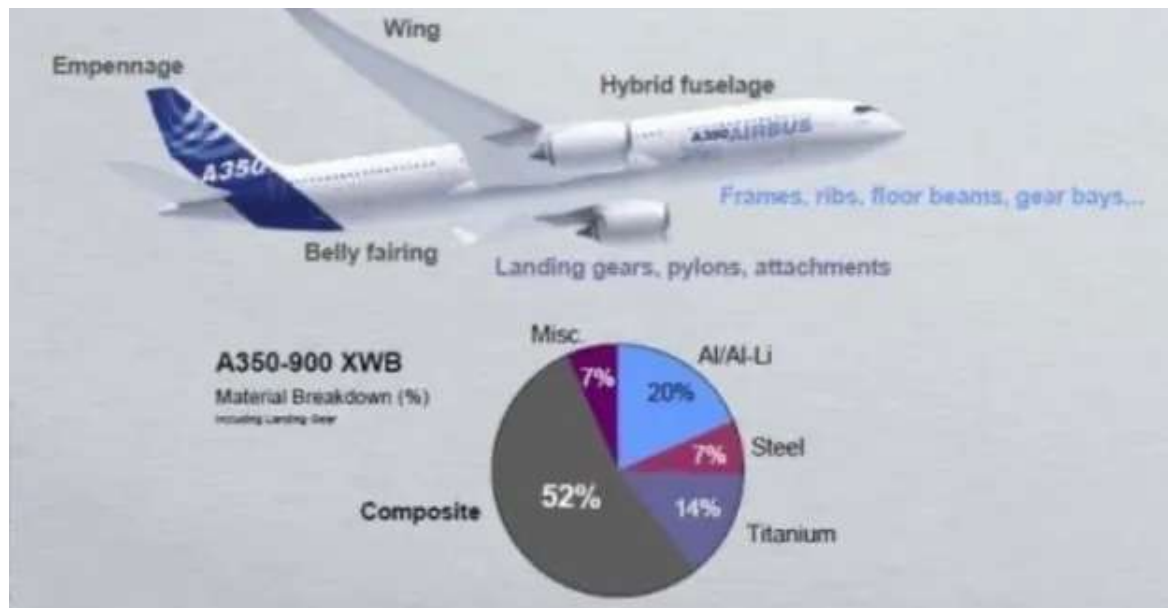
Can reduce Lift

Enough to balance the reduced weight

For each kg reduction of aircraft mass, about 372 kg less CO_2 is produced every year

More weight efficient airframe





A380 | OEW/MTOW ~ 0.54

A350 | OEW/MTOW ~ 0.45

Recycling steps

- Step-1: Remove all hazardous materials
  - Fuel, hydraulic oil, waste-water etc.
- Step-2: Disassemble valuable parts (~50% wt.), repair (if needed) and re-use in the aviation market (components for re-use)
  - Undercarriage, engines, avionics, seats etc.
- Step-3: Dismantle parts (~ 35-40% wt.) and recycle for non-aviation market (materials for re-cycling / energy recovery)
  - Metallic parts, plastic interiors etc.)
- Step-4: Incinerate / landfill with non-recyclable wastes ~~(~10-15% wt.)~~ (~ 50-60% wt.)
  - Carbon composites, insulation blankets, carpets, cushions, flame retardant panels etc.

AIR-International-Dec-21-aircraft-disassembly.pdf (wingborn.com)

Reducing weight

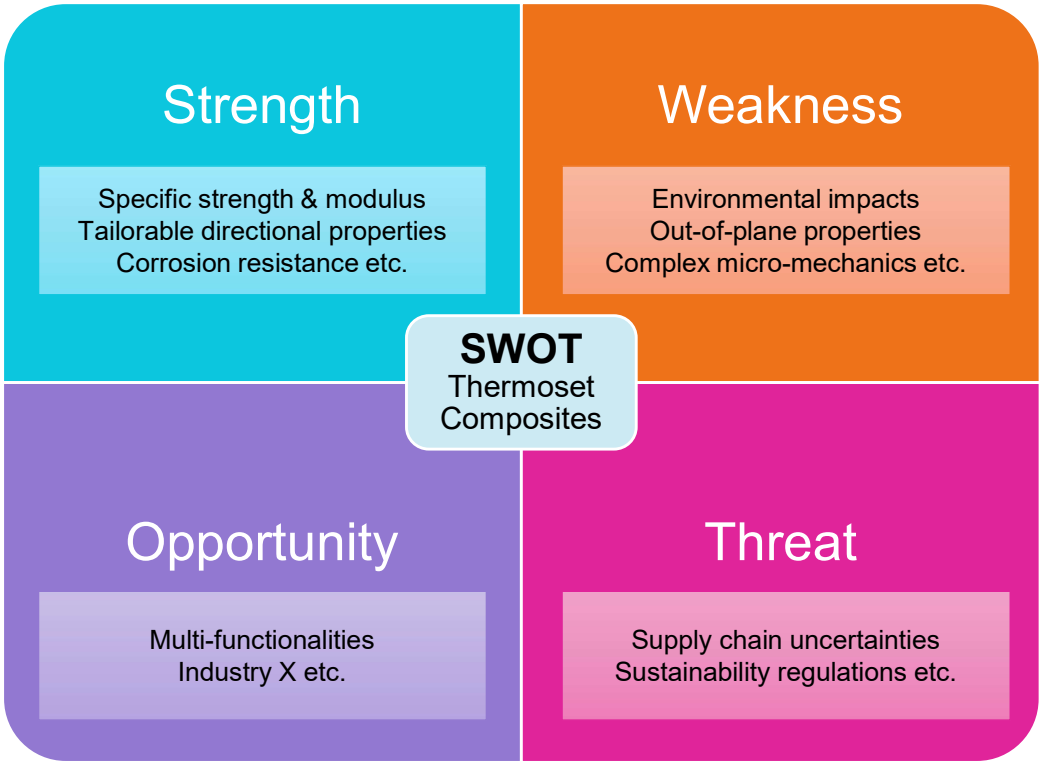
- Novel structural concepts (truss based etc.)

- Reduced FoS
  - Better load estimation
  - Better manufacturing control

- Structural Health Monitoring (SHM) for need based maintenance

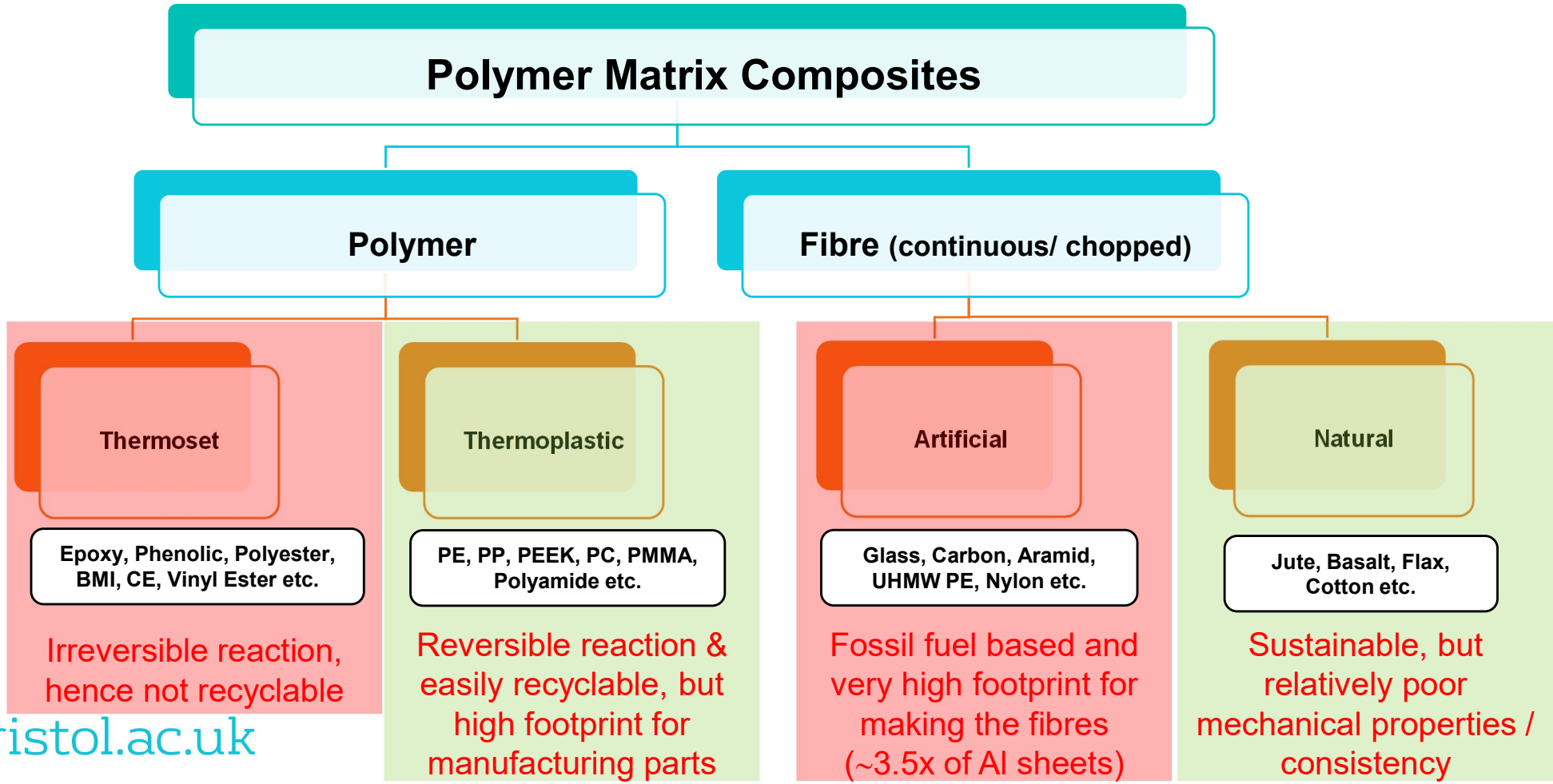
- **<u>More composites?</u>**

More composites?



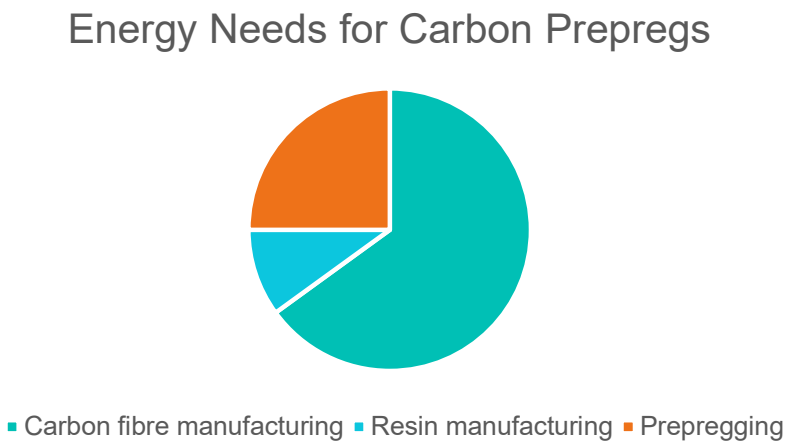
- ✓ **Eco-friendliness**
  - TP composites
  - OoA processes
  - Recycled fibers
- ✓ **Out-of-plane prop.**
  - 3D composites (for more durability & damage tolerance)
  - Matrix mods
  - Self healing
- ✓ **Complex μ-m/ics**
  - Online SHM
  - Digital Twins
- ✓ **Supply chain**
  - Distributed manufacturing @ Industry X
  - Alternates to prepregs
  - C Fibre capacity expansions

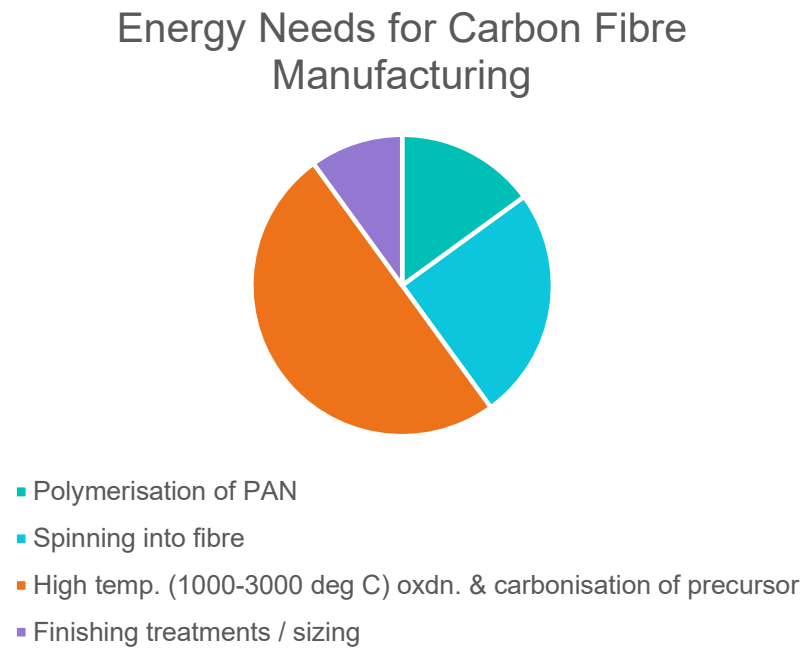
The raw materials



Making of carbon fibers

- Oxidative stabilization
  - PAN precursor is stretched & oxidized in a temperature range of 200-300°C.
  - Converts thermoplastic PAN to a non-plastic cyclic or ladder compound.
- Carbonization
  - Fibers are carbonized ~1000°C w/o tension in inert atm. (normally N_2) for few hours.
  - Non-carbon elements are removed as volatiles (yield of about 50% of PAN)
- Graphitization
  - Fibers are treated at temperatures between 1000-3000°C, to improve crystallinity.





Room for Energy Optimisation

- Carbon fibre manufacturing process (max contributor)
  - Short term
    - More efficient furnaces
    - Waste energy recovery
  - Long term
    - Targeted heating (microwave / induction)
    - Recycling carbon fibres as precursors (circularity)
    - Less energy intensive precursors like polyolefins
- Other possibilities
  - Alternate OoA processes (RTM) without prepregs
  - Alternate resin systems like thermoplastics
  - Distributed manufacturing & localised supply chains
  - Recycled carbon fibres

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OoA Processes

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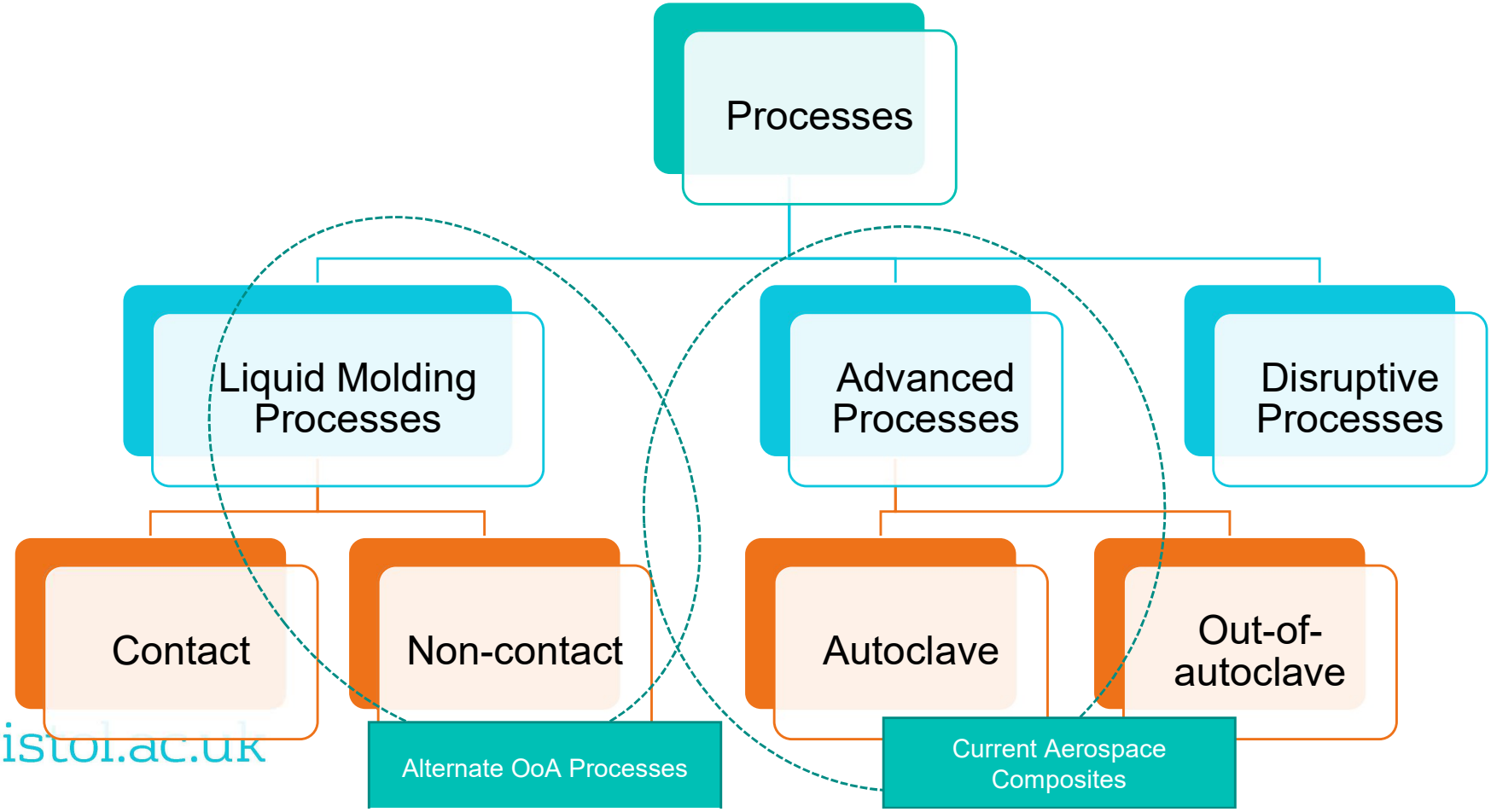
Thermoset manufacturing processes



Conventional liquid molding

Hand Lay-up

- Only hand pressure during the lay-up
- Low Vol. fraction (<40%) & mechanical properties
- Very cheap

Vacuum Bagging

- Hand lay-up followed by pressurization through creation of vacuum within a flexible membrane stretched
- %) and mechanical properties
- Minor cost additions (equipment and consumables)

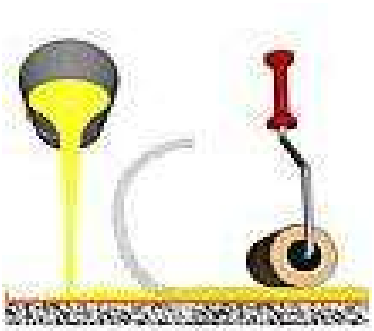
Matched Die Molding

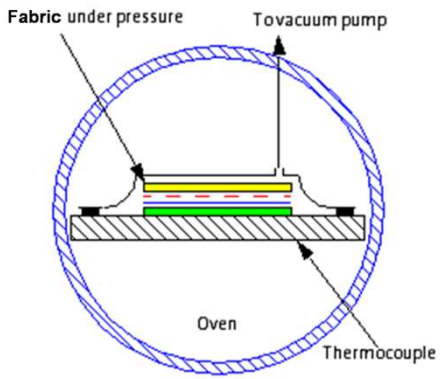
- Hand Lay-up followed by pressurization through compressive forces by closing the male-female mold
- Improved control over thickness
- Major cost of molding tools (if metallic)

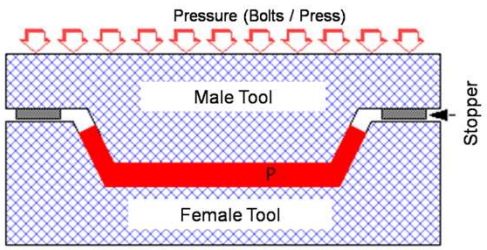
Other liquid molding processes

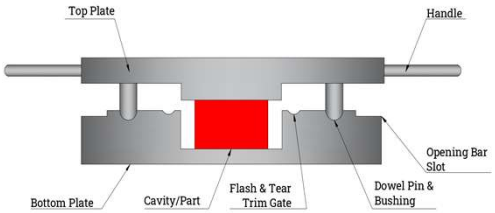
- Compression molding
- Pultrusion etc.

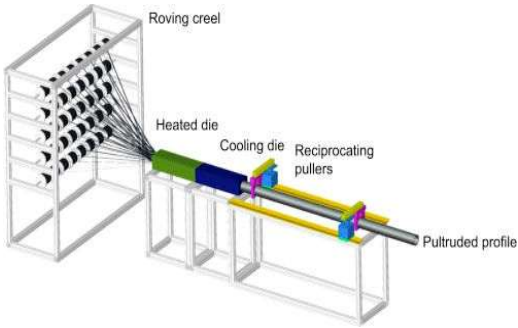
Generally unsuitable for safety critical applications



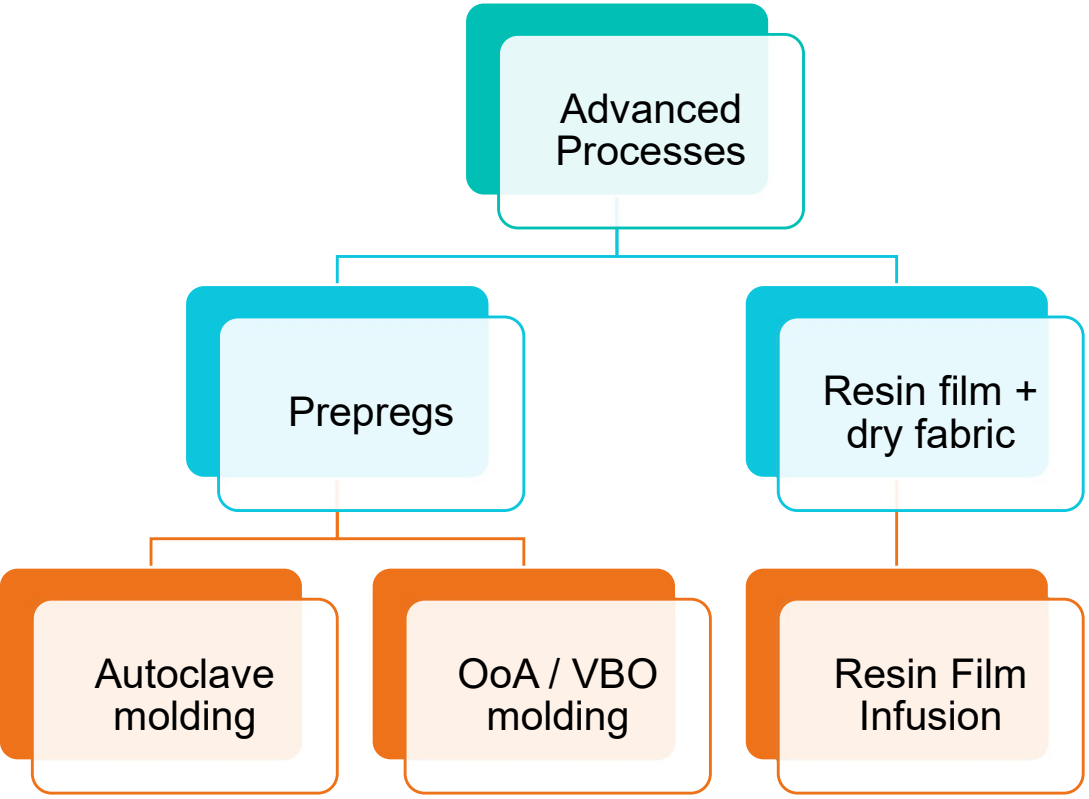








Advanced processes



Prepreg Molding (Autoclave / OoA)

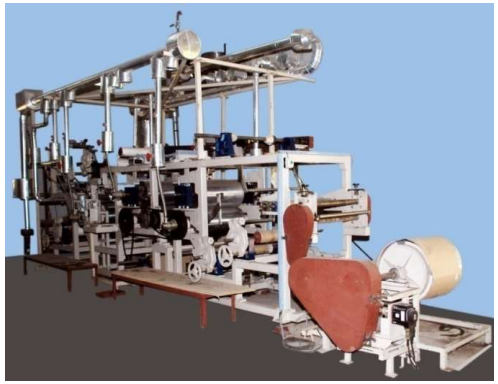
- Semi-cured (B-stage) resin impregnated fabrics laid inside mold

Generally used in aerospace

- Excellent thickness control and repeatability
- Huge cost (capital, prepreg raw materials & electricity)

Resin Film Infusion (RFI)

- B-stage cured resin films made in calendaring machine
- The films sandwiched in between dry fabrics and cured under high temperature and vacuum
- Relatively high vol. fractions (55-60%) and mechanical properties
- Reasonably good thickness control and repeatability
- Optimal cost, less shelf life dependency



Non-contact liquid molding

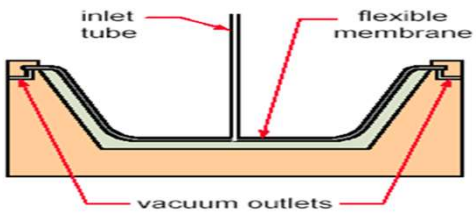
RIM / VARIM

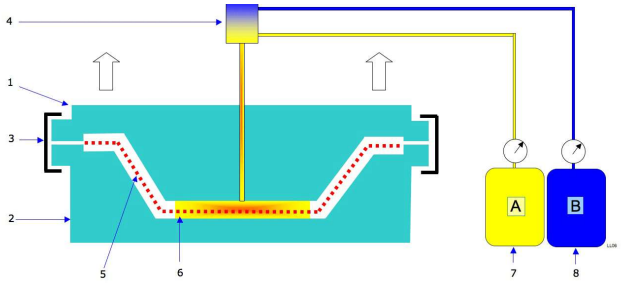
- Dry preforms positioned inside the female mold cavity
- Resin sucked in through vacuum
- Pressure applied almost 1 bar
- Moderate vol. fraction (45-55%) and mechanical properties
- Non-contact and worker friendly
- Almost automated lay-up process
- Easy positioning of inserts
- Part consolidation
- Extensive process development for standardization and optimization required

RTM

- Resin is injected under very high pressure inside the mold cavity pre-laid with dry preforms
- Less s ation as co
- Chances of fiber shifting under high pressure
- Moderate vol. fractions (45-55%) and mechanical properties
- Extensive process standardization required

Why not?





Cost & weight comparisons (HS carbon-epoxy)

Process	V'f (%)	V'o (%)	t (mm)	m (Kgs)	Costs per component (US $)				
					Feedstock	Tooling	Manpower	Plant	**Total**
HL	40	10-25	3.7	7.1	800	80	800	10	**1690**
VB	45	5-10	3.3	6.3	850	80	800	130	**1860**
RIM	50	3-6	3.0	5.8	850	100	800	130	**1880**
RTM	50	2-5	3.0	5.8	850	160	800	130	**1940**
Autoclave	60	0-1	2.5	4.84	1600	200	1200	1000	**4000**

- *Data partly taken & estimated for present values from Michael G Bader ; Composites Part-A 33 (2002) 913 – 934*
- *Autoclave process is almost 2-2.5 times more expensive w.r.t the out-of-autoclave processes*
- *Equivalent autoclave component is about 15-30% lighter than the liquid molded one*

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Thermoplastics

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Alternate resin systems: Thermoplastics

Poly –ether-imide (PEI) – Amorphous | Nylons – Amorphous or Semi-Crystalline | **Poly-ether-ketone-ketone (PEKK) – Semi-Crystalline** | **Poly-ether-ether-ketone (PEEK) – Semi-Crystalline** | Poly-phenylene-sulfide (PPS) – Semi-Crystalline

Sustainability comparisons

Structural Composites	Life Cycle Assessment			
	Environment impacts -Global warming -Human toxicity -Air pollution	**Used resources** -Coal -Other fossil fuels -Water	**Waste types** -Hazardous -Nuclear	**Output flows** -Re-usable -Re-cyclable -Incineration/ landfill
Thermoset Carbon Composites	•Large C02 eqv. •Toxic & inflammable	•Petroleum precursors •Moderate power	Hazardous	Incineration / landfill
Thermoplastic Carbon Composites	•Large C02 eqv. •Toxic **(but less production)**	•Petroleum precursors **•Huge power**	Hazardous **(but less waste)**	**Resin: Re-cyclable** **Fibre: Re-usable**

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Carbon Fibre Recycling

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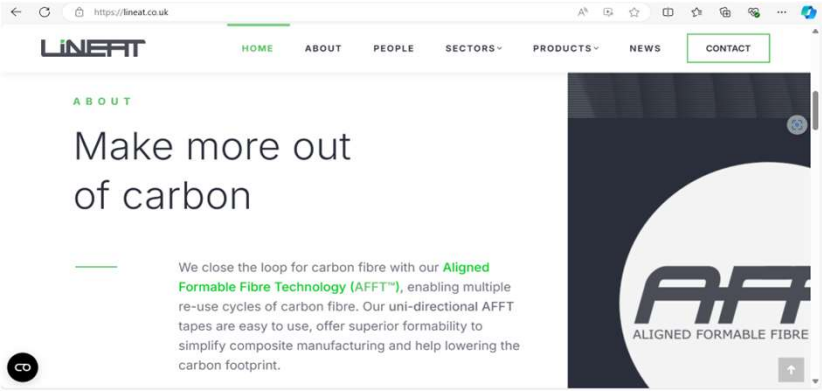
Aligning recycled carbon fibres



Technical Data Sheet

LINEAT

Lineat Composites
Unit 3 Severn Link Distribution Centre
Newhouse Farm Industrial Estate
Chepstow NP16 6UN

AFFT™ ALIGNED FORMABLE FIBRE TECHNOLOGY

AFFT™ tapes are advanced uni-directional pre-preg tapes made with highly aligned short fibres. The highly aligned fibre architecture mimics virgin continuous fibre material and it can reach similar performance as equivalent continuous fibre materials, whilst adding unique moulding and draping characteristics to speed up manual and automated lay-up procedures. Tapes with custom (reclaimed) carbon fibres can be made to create strong recycled composite tapes that can substitute continuous fibre UD material.

OVERVIEW

Fibre type	rCF: Reclaimed standard modulus 4mm carbon fibre
Fibre aerial weight	33 gsm (typically +-3 gsm)
Resin system	Epoxy (SHD MTC400-1)
Resin aerial weight	50 gsm (typically +-10gsm)
Tape width	100 mm
Cured ply thickness	0.03 mm (typical)

PROCESSING

Method	Vacuum / oven cure	Autoclave	Compression moulding
Consolidation pressure	1 bar	7 bar	10 bar+
Fibre volume fraction V_f	20 – 30 %	30 – 40 %	40 – 50 %

STORE FROZEN AT -18°C.
Cure temperature-time*:
85°C – 16 hours
100 °C – 4 hours
135 °C - 1 hour

**Please see resin TDS for more detailed cure information.*

TYPICAL COMPOSITE PROPERTIES

Processing conditions: Compression moulding at 90 bar. Estimated FVF of 46%.

Property	Unit	Value	Standard
Tensile Modulus	GPa	89.40	ISO 524-5
Tensile Strength	MPa	722.76	ISO 524-5
Flexural Modulus	GPa	89.40	ISO 14125
Flexural Strength	MPa	967.78	ISO 14125
Compressive Modulus	GPa	92.31	ISO 14126
Compressive Strength	MPa	928.03	ISO 14126
ILSS	MPa	89.17	ISO 14130
Density	g/cm^3	1.45	-

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3d Composites

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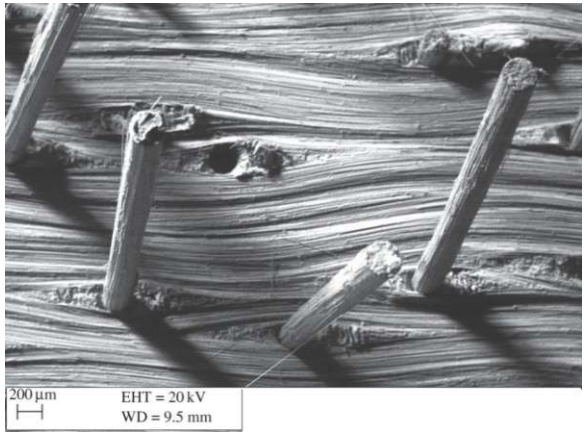
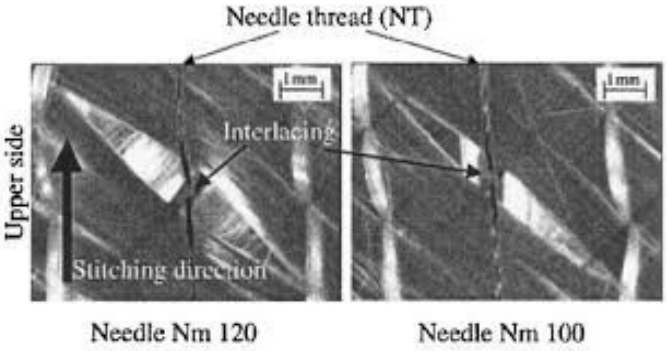
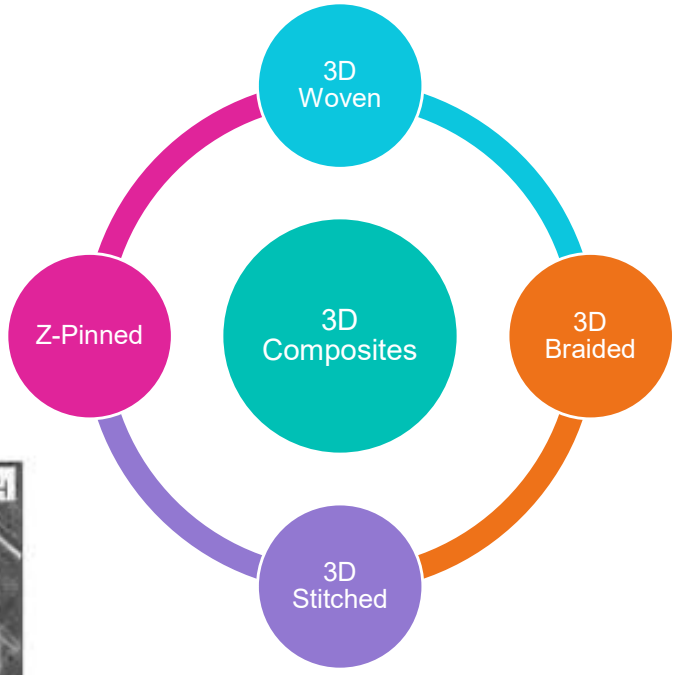
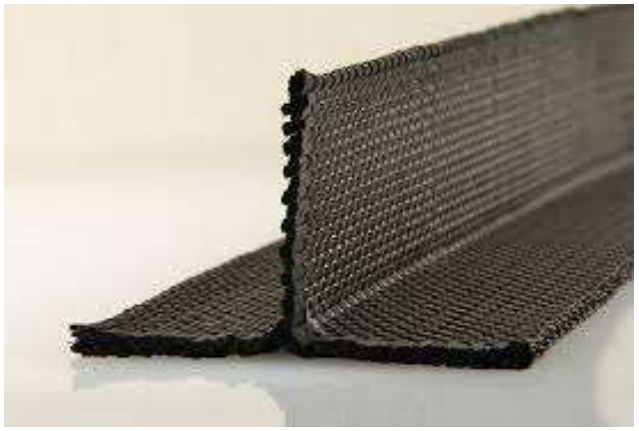
Commonly used 3d Composites









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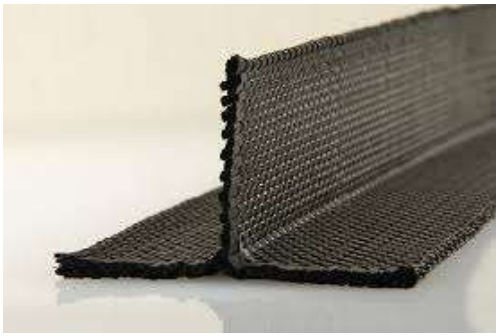
3D Woven Composites

- The 3-D woven fabric is a variant of the 2D weaving process
  - Multiple layers are woven together to produce 3 dimensional woven fabrics
  - Could be formed to near net shape with considerable thickness up to 10cm
  - No need for layering to create
  - Fibers placed in the thickness direction are called z-yarn, warp weaver, or binder yarn, which interlaces warp and weft yarns of different layers
- 3d woven structures can create composite materials with fiber volume fractions around 50% (generally through RTM process)
- Can be created into nearly endless shapes (standard I-Beam, sine-curve I-Beam, airfoils etc.)
- Used for various aerospace applications
  - Engine casings and fan blades for A-320-neo, 737-MAX and Comac C-919
  - Engine rotors & mounts
  - T- and X-shape panels, leading edges for aircraft wings, I-Beams etc.



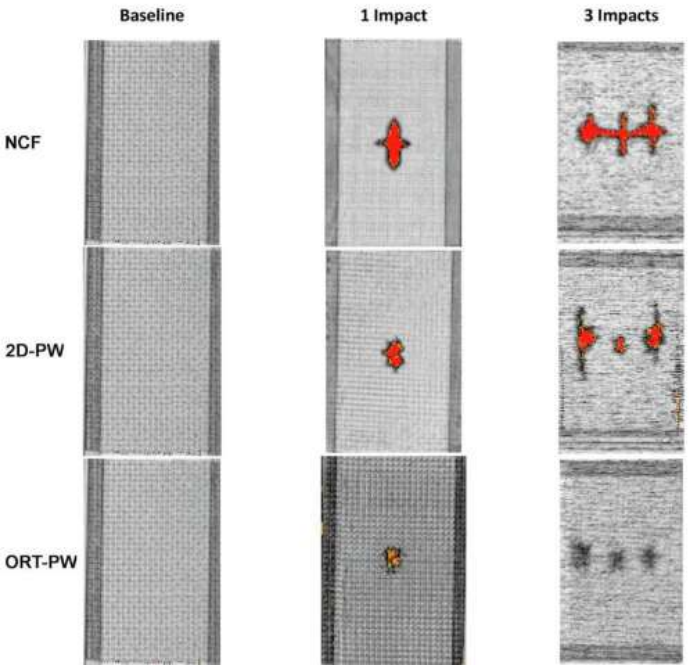
Advantages of 3D Woven Composites

- Out-of-plane properties (reduced delamination)
- Formability (for complex composite designs)
- Reduced takt time
  - Near net shape fabrication (no need of multiple layers)
  - Reduced infusion time (due to higher porosity in fabric)
- Z-directional thermal conductivity

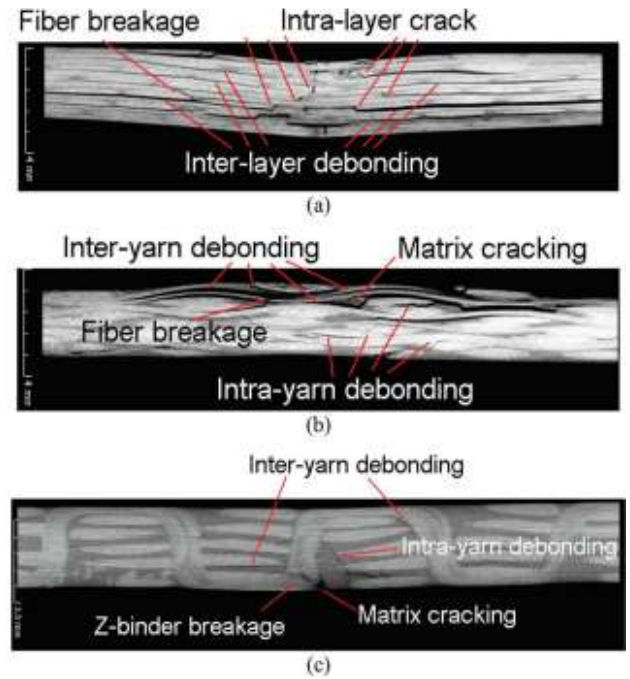
Properties of 3D Composites

- In-plane properties (tensile, compressive, shear etc.)
  - Various types of defects are inadvertently created during the 3D weaving / braiding / stitching / pinning process
  - Similar or slightly higher than an equivalent two-dimensional (2D) material
- Out-of-plane properties (CAI, bird strike, FOD etc.)
  - Significantly improved, with minimal compromise in in-plane properties
    - Damage assessment of NCF, 2D and 3D woven composites under compression after multiple-impact using C-scan, X-ray, CAI & acoustic emission
    - By Milad Saeedifara et al | Recent publication (2020) | Composites Part-A (Elsevier)
    - https://eprints.whiterose.ac.uk/157302/8/1-s2.0-S1359835X20300713-main.pdf

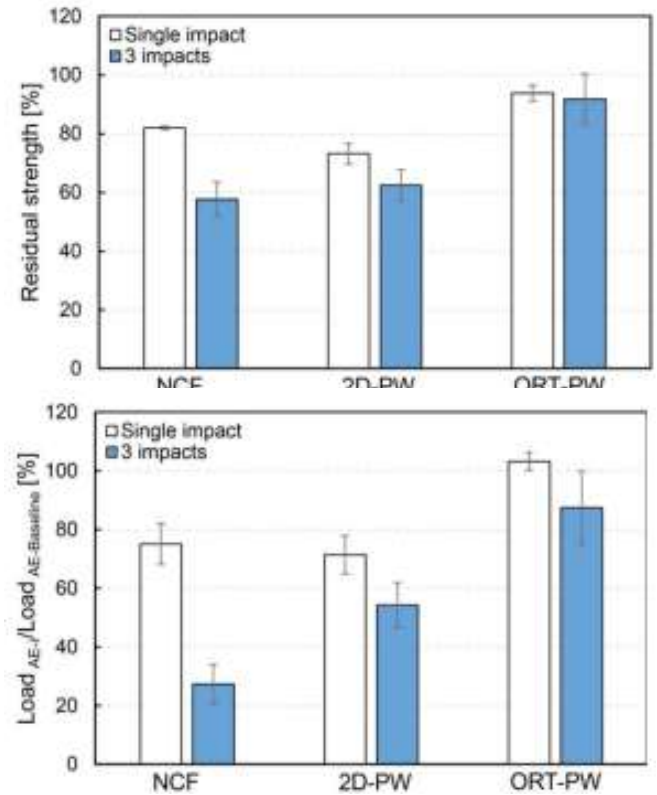
Properties of 3D Composites (contd.)



C-scans after 15J Impacts



X-Ray CT after 1st Impact



Damage Progression

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Distributed Manufacturing

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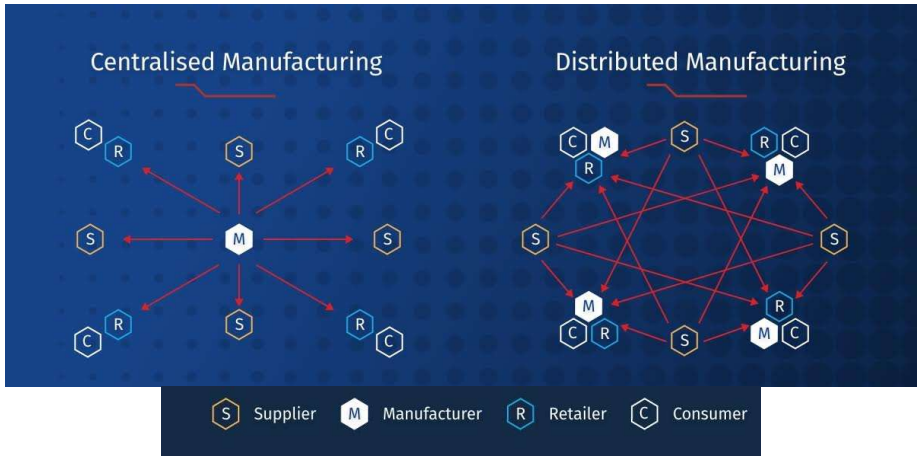
Decarbonisation scopes in aviation

- Scope 1: Direct emissions from aircraft engines, ground support equipment, vehicles owned by the airline etc.
- Scope 2: Purchased electricity, heat, or steam etc.
- **Scope 3:**
  - Fuel & Energy Related Activities not included in scope 1 or 2
  - **Purchased Goods / Parts** and Services (extraction, production, transportation etc.)
  - Business Travel
  - Leased Spaces in airports

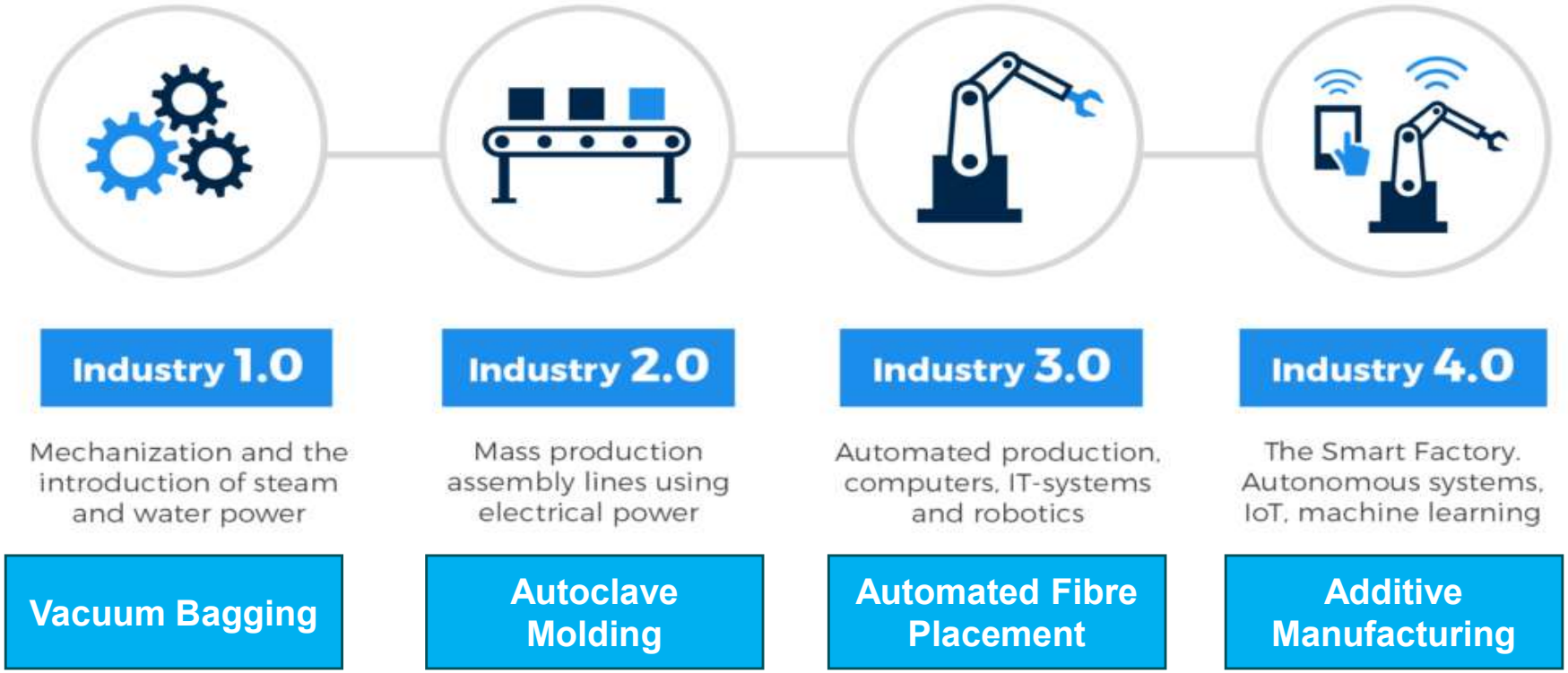
Decarbonizing aerospace | Roland Berger

Distributed manufacturing & supply chain

- Distributed manufacturing (aka hyper-localised manufacturing) aims to bring manufacturing closer to the customer through setting up distributed facilities across the globe
- The concept was born due to the ever-rising fuel & logistics prices and **environmental impacts**
- Additive manufacturing (AM) technology is a primary enabler for distributed manufacturing - distributed AM is the use of AM for the transition to distributed manufacturing
  - Rapid on-demand low volume manufacturing (less cost on warehousing & transportation)
  - Digital inventory (no stocking, only sending digital file to local hub)
  - N=1 experience (mass customisation)
- Few challenges
  - Digital talent sourcing
  - Cyber security
  - IPR in distributed manufacturing

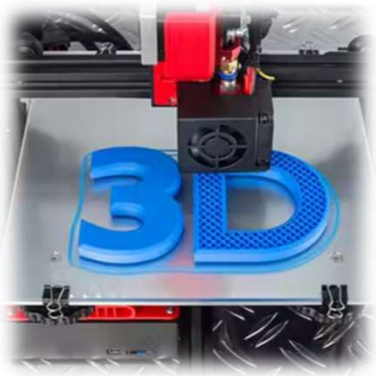


Evolution of composite processes



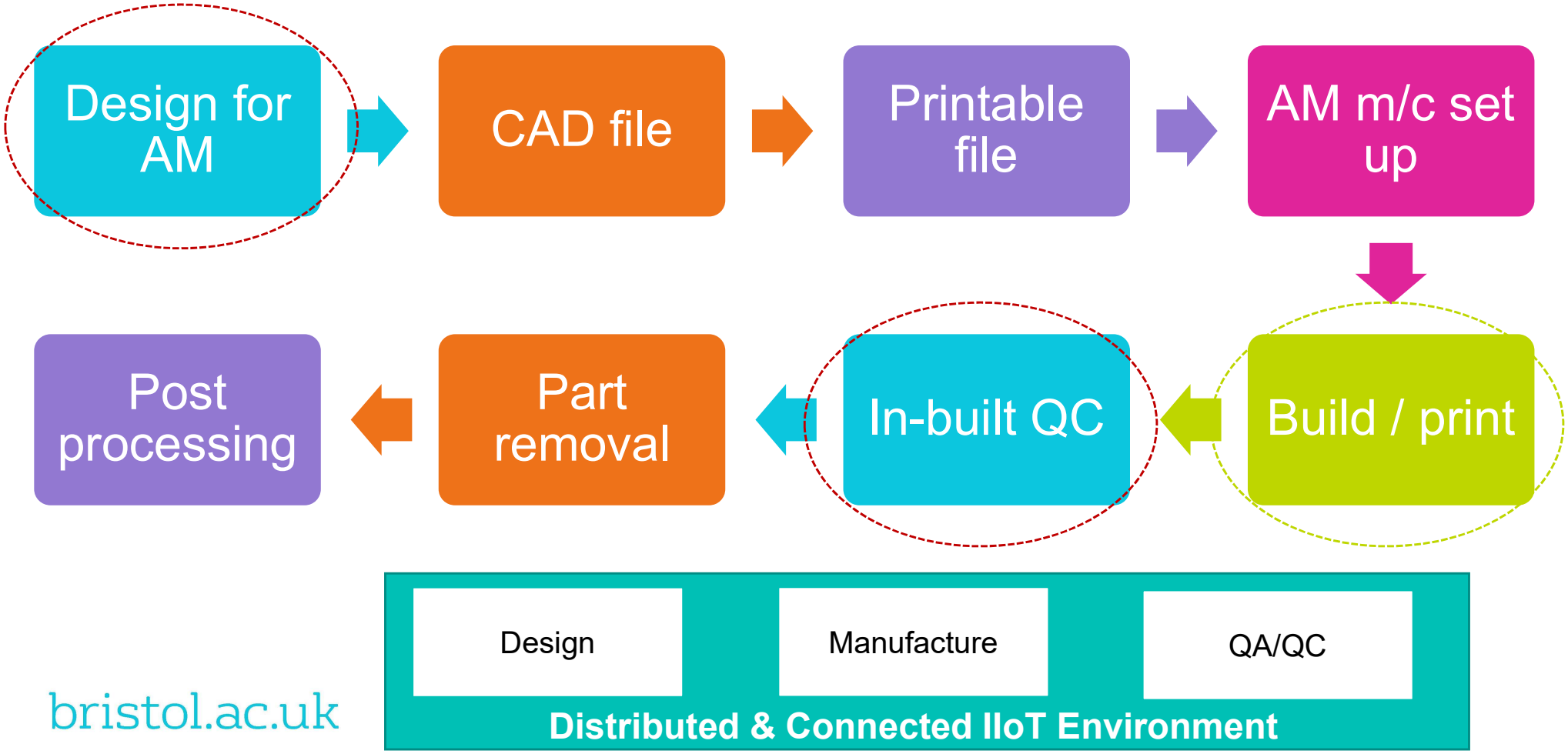
Additive manufacturing

- Traditional understanding = 3d printing
  - Bottom up building, layer by layer (as against subtractive manufacturing)
    - Ancient example: Egyptian pyramids
  - Minimize wastage
    - Material cost↓
    - Energy cost↓
    - Storage cost↓
    - Environment friendliness↑



- Umbrella concept
  - Transformative approach in industrial products
  - An essential pillar of IIoT / Industry 4.0 & X
  - Traditional understanding (i.e. 3d printing) is only a part of it

Umbrella concept



AM in different industry

Automotive

- RPT
- New designs
- N=1
- Less waste (environment)
- Cost & weight reduction
- Example: Sports car frame

Healthcare

- 3d scanning & reverse engineering
- N=1
- Biocompatible materials
- Example: Hip implant

FMCG

- Impossible design concepts
- Accelerated market entry &
- Agile product customization
- Example: Customized shoe soles

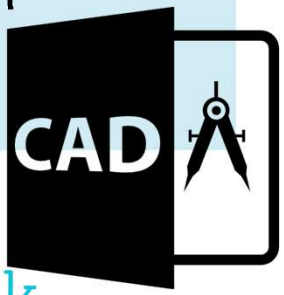
A&D

- Weight saving
- Complex geometry
- Low volume
- Example: Small & complex engine parts, brackets, spares etc.

Why AM?

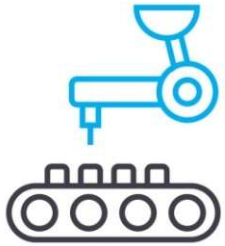
Design

- Part consolidation
- Weight savings
- Complex parts by design optimization (material / geometry)
- ML in design optimization



Manufacture

- Rapid prototyping
- JIT manufacturing
- **<u>Distributed manufacturing</u>**
- Reduction in inventory
- Reduction in scrap
- Reduction in cost
- N=1 experience

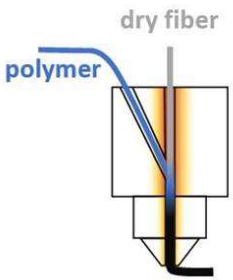


Quality

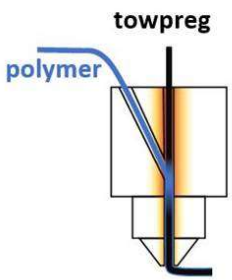
- Consistency (lesser variance)
- Real time quality inspection
- Automation of QC

AM of continuous fibre composites

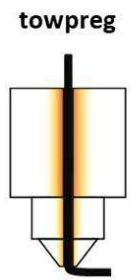
In-situ Impregnation



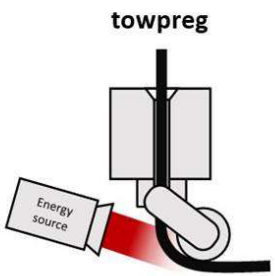
Co-Extrusion with Towpreg



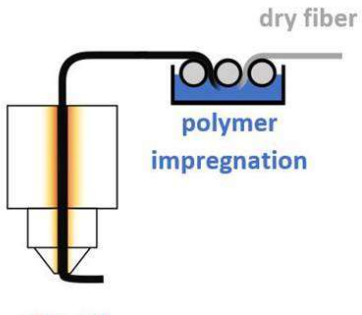
Towpreg Extrusion

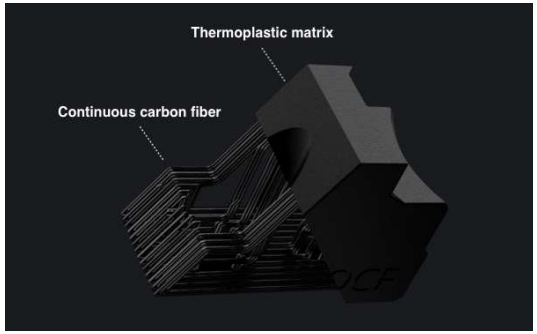


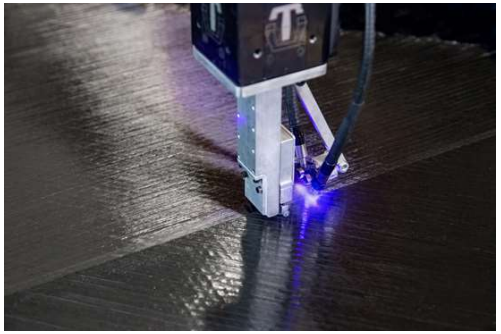
In-situ Consolidation



Inline Impregnation





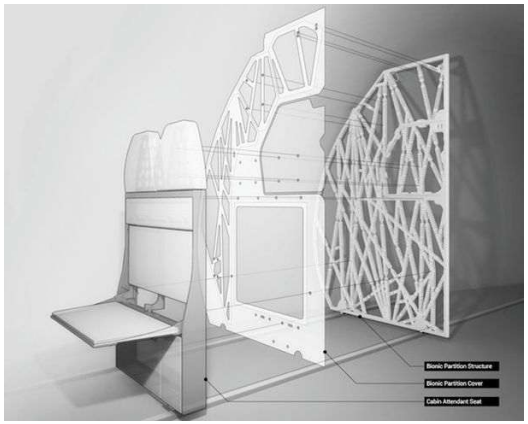


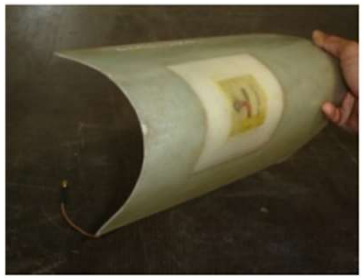
Green & sustainable composite structures for aviation

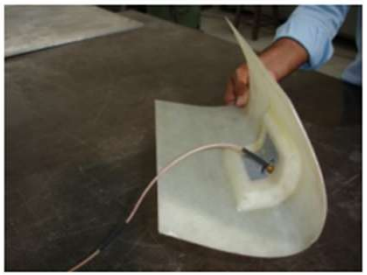
Materials	**Matrix:** Thermoplastic (high performance PEEK / PAEK etc.) **Reinforcement:** • Carbon fibres from recycled precursor (circular economy) • Recycled aligned carbon fibres • 3D reinforcements for out-of-plane durability
Process	Low T&P; minimize energy wastage, use green energy • OOA processes like RTM • Additive manufacturing • Self heating molds / microwave curing etc.
Supply Chain	Hyper-localized (IIoT enabled distributed AM)
Structures	Sustainable structural design concepts

Sustainable structural design concepts

- Optimised structural frameworks
  - Truss based wings – Boeing / NASA X-66
  - Biomimetic lattice structure – Airbus Bionic Partition
- Multi-functional load-bearing structures
  - Energy storing carbon electrodes for structural battery - Chalmers University
  - Heat sink wings to reduce cooling requirements form electric motors - MIT
  - Conformal load-bearing antenna structure for drag reduction – Military aircraft / UAVs
- Adaptive load distribution systems
  - Active aeroelastic wings to distribute loads - Boeing X-53
  - Morphing wing with interlocking carbon fibre elements - NASA MADCAT







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Thank you

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